Bocconi

BUSINESS STRATEGY
30060 – MANAGEMENT

Appendix

Università Bocconi
MILANO

Average unitary costs (AUC) = total production costs/actual quantity produced

We can explain differences in AUC considering:

Scale economies

- Scale economies: lower average unitary costs that may result from increasing MPC while keeping the degree of utilization constant
  - Imagine a plant that operates at MPC=x and we increase it up to y maintaining the degree of utilization constant
  - We achieve scale economies if AUC(y)<AUC(x)

Economies of fixed cost absorption

- Reduction in AUC obtained by dividing fixed costs across larger units of production output
  - We realize fixed-cost absorption economies by increasing the rate of utilization for a given MPC
  - Greater benefits in industries where fixed costs represent a higher fraction of total costs: medical sector , education, services and manufacturing to some extent

Coherently with the definition, to quantify scale economies we compare AUC of the investment alternatives at the same production capacity.

Scale economies: $\boxed{AUC_A - AUC_B}$ (at the same production capacity)

We compute the AUC of B at the same production capacity using the formula:

$$\boxed{\frac{Fixed\ costs}{Maximum\ capacity\ of\ B\ \times CPC\ of\ A} + Variable\ cost\ per\ unit\ (B)}$$

In the same way, to quantify economies of FCA, we need to compare the AUC at different level of production capacity.

Economies of fixed cost absorption

$$\boxed{AUC_{B\ at\ x\%} - AUC_{B\ at\ y\%}}$$